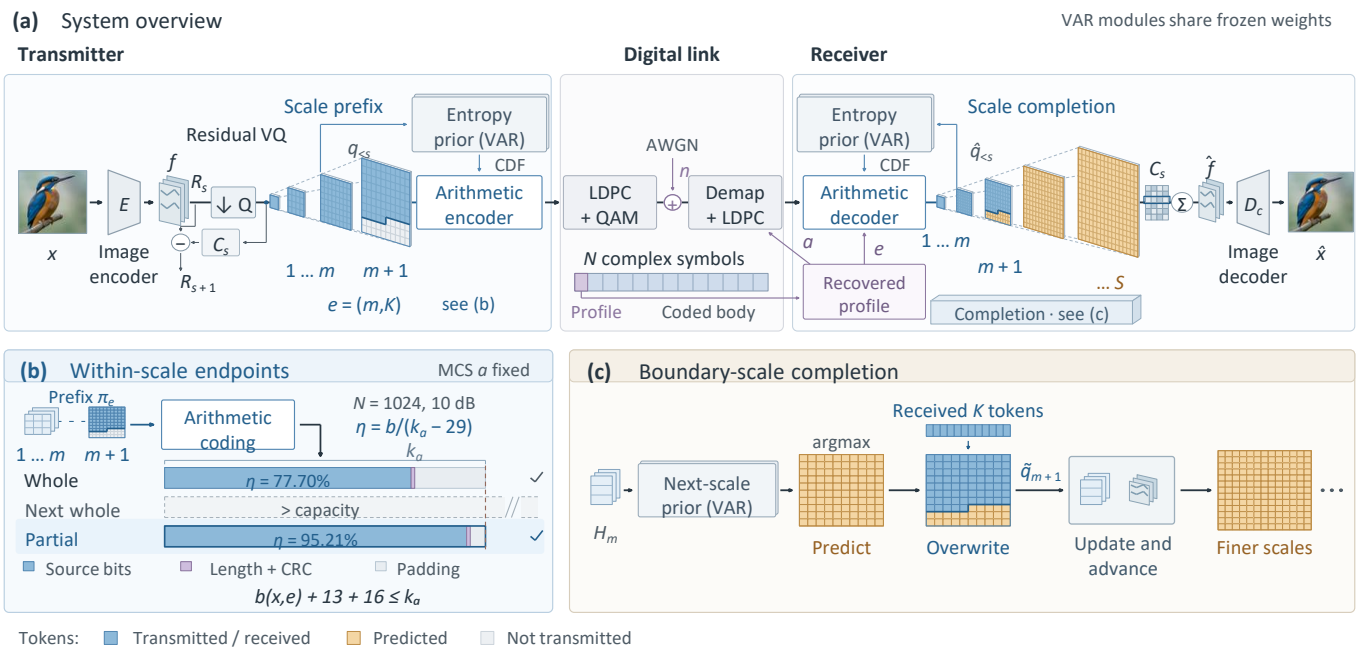

(a) System overview
VAR modules share frozen weights
Transmitter
Digital link
Receiver
Scale prefix
Entropy prior (VAR)
Residual VQ
CDF
AWGN
Entropy prior (VAR)
Scale completion
CDF
Arithmetic encoder
LDPC + QAM
Demap + LDPC
Arithmetic decoder
Image encoder
Image decoder
1 … m
m + 1
N complex symbols
Profile
Coded body
Recovered profile
… S
e = (m,K)
see (b)
Completion · see (c)
(b) Within-scale endpoints
MCS a fixed
Prefix π_e
Arithmetic coding
N = 1024, 10 dB
η = b/(k_a − 29)
Whole
Next whole
Partial
η = 77.70%
> capacity
η = 95.21%
Source bits
Length + CRC
Padding
b(x,e) + 13 + 16 ≤ k_a
(c) Boundary-scale completion
Received K tokens
argmax
Next-scale prior (VAR)
Predict
Overwrite
Update and advance
Finer scales
Tokens:
Transmitted / received
Predicted
Not transmitted